Title: IoT-Smart Management System for an efficient Football Pitch

line 1: Jawher Ftiti
line 2: *Computer Science*
line 3: *name of organization (of Affiliation)* OTH Regensburg
line 4: email address
jawherftiti26@gmail.com

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Keywords—component, formatting, style, styling, insert (*key words****)***

I. INTRODUCTION (*HEADING 1*)

Here we would like to present the design and implementation of an Internet of Things (IoT) System for an efficient management of football fields. The system integrates various IoT devices to monitor and optimize field conditions, enhance player safety, and enable also a good maintenance operation. So we will also discuss about the purpose of the system, describe its key components and functionalities, and also give some potential use cases in order to improve the overall football field management experience.

II. DESCRIPTION OF THE IOT SYSTEM

No one can deny that football has been since long time one of the most popular sports globally, so it also requests well-cared-for playing surfaces for player safety and performance. The Internet of Things (IoT) based football field management system is a revolutionary solution that focuses on improving pitch conditions, maintaining the player safety and simplify maintenance tasks. The system consists of a series of IoT devices (sensors and actuators) interconnected to a centralized platform that gathers, analyzes and processes the information in real-time. The system's architecture is designed for efficiency, using an edge computing architecture to minimize latency, cut down on data transmission costs, and boost the control of field and player management operations .The main objective of the system is to collect real-time data of several parameters (e.g., turf moisture, turf temperature, turf hardness, and player activity…), analyze this data to detect trends and anomalies, and act proactively to maintain optimal playing conditions.

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Fig. 1. Conceptual image of a smart management system for an efficient football pitch .

III. Main Components of the(IOT) and an Overview of the System Design

1. **Sensor Network:** The system utilizes a network of sensors deployed across the football field to neters such as turf moisture, temperature, hardness, and ambient weather conditions

(rainfall, sunlight) . These sensors continuously collect data and transmit it to a central hub or cloudbased platform for analysis. At that point, the system uses a variety of sensors to collect real-time data on the football pitch and the players such as Turf Sensors for example that aims to control the football field conditions and remains it playable for the players. These types of sensors collect the temperature of the soil, soil moisture and its pH level. The system contains also weather Sensors that also detect the environmental conditions that might affect the pitch and the players like temperature , Humidity , wind Speed and directions and Rainfall levels . there are also the Player Wearables that ought to track the players Performance and Health such as Heart rate , GPS coordinates and injury detection .

2. **Automated Maintenance and Control:** A trigger status from the analysis results would then tell the system how and when to perform the relevant automated maintenance actions — like scheduling irrigation cycles, turf aeration, and field-marking to optimize playing conditions and player safety.

3. **Player Performance Monitoring:** In addition to Field conditions, the system also monitors the body performance ability and out of the box accounting chips . By translating This into data, the wear and tear of players on the pitch will be readily available so that coaches and medical staff can predict workloads, fatigue and an injury risk and make suitable training units and rotations.

4. **Cloud Database and Analytics**: The sensor data is stored in the centralized cloud database. This data powers real-time analytics, historical trend identification, and action-based intelligence. “Then there are the advanced analytics of the cloud that can help predict trends and assist with decisions on pitch management or player health.

By implementation of these technologies we provide IoT-Smart Football Pitch Management System which helps to maintain condition of football pitch and determine player health. The football pitch and the players provide real time data collected by a network of sensors and wearables. This data is then analysed and used to trigger automated actions for maintenance as well as provide actionable insights to coaches, medical staff, and groundskeepers.

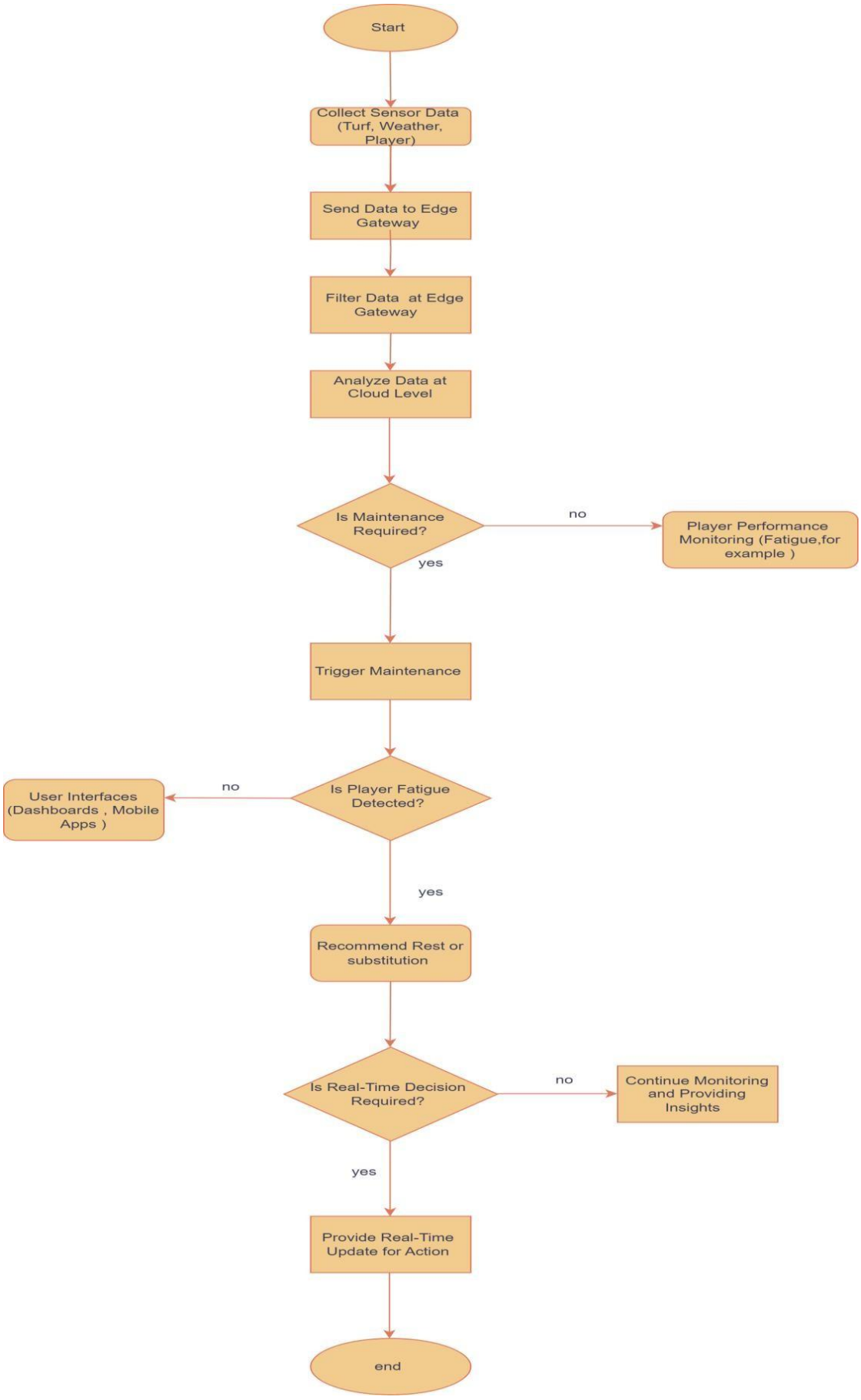


Fig 2 : The Flow Diagram of the IOT Management System for a football Pitch .

The flow diagram visually represents the sequential process of the IoT system from data collection to decision-making. The process begins with the collection of sensor data (such as turf conditions, weather, and player health metrics) and proceeds through the edge gateway for initial data processing.

The data is then analyzed for greater insights, and predictive analytics. Using this analysis, maintenance actions (e.g., irrigation or aeration) are initiated as needed, as well as continuous

monitoring of player performance to identify the onset of fatigue and risk of injury.
Maintenance procedures are designed to be run only when necessary (especially in the case of unexpected failures), and triggers are monitored so that unnecessary maintenance does not get carried out (unless the system has been used to run the monitoring procedures for any faults), hence, the flow diagram includes decision points that ensure that resources are used effectively; the system also outputs an interface for the stakeholders to visualize insights in near real-time. The cycle of this process is repeated endlessly in order to create ideal pitch conditions and player Health . The flow Diagram is ideal and optimal for the IoT-Smart Football Pitch Management System it simplifies the complexity by making the system more easily understood as a diagram showing how all the different components (sensors, edge gateways, cloud analytics) interconnect and interact easily . This diagram depicts also logical flow of operations, from data collection to decision-making, creating transparency in the processes that the system employs. Moreover it shows key decision-making steps (e.g., “Is maintenance required?”), making resource-efficient logic and automated actions easy to grasp, enhances the communication between technical teams and non-technical stakeholders like coaches and groundskeepers and it emphasizes also feedback loops and conditions for triggering actions, demonstrating how the system conserves resources like water and energy.

How would The Field Maintenance Decision be made?

The system monitors parameters like soil moisture, turf hardness, temperature. Sensors collect data at regular intervals, so for example here we can say at every 10 minutes, ensuring real-time control.
Maintenance Decision are predefined, for example, triggering irrigation when soil moisture falls below 30%.
The system aggregates sensor data and analyzes it using edge gateways for quick, localized processing. Predictive analytics use historical trends, allowing us for proactive maintenance before conditions getting worse .

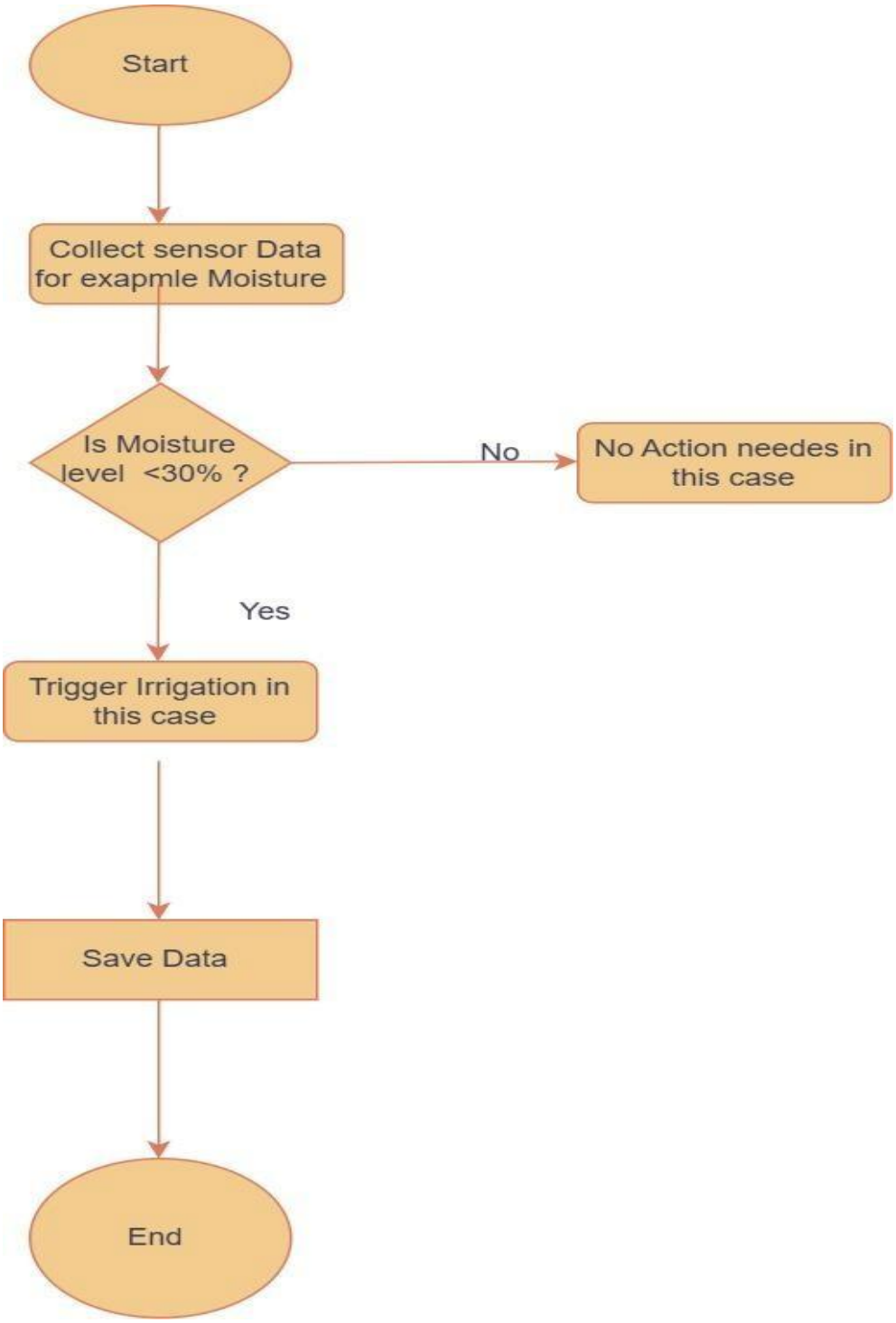


Fig 3 : A Simple Field Maintenance Flow Diagram

This diagram shows how the system monitors the field. Sensors deployed across the field collect data like moisture Level and temperature for example regularly. If the data exceeds a set limit (moisture < 30%), the system activates maintenance (like irrigation). If not, it keeps monitoring and saves the data for later use.

Details about the Player Monitoring Process :

What kind of sensors are used for the Player Monitoring Process?

Wearable devices, heart rate monitors (For the workload), and GPS trackers (for the location) are used. These sensors collect and transmit data every second to provide an overview of player performance.

Heart rate data determines exertion levels, while wearable devices can identify signs of fatigue. And then this data would be analysed to assess injury risks and recommend recovery periods, ensuring players remain in optimal condition.

The real time feedback of the coaches or Medical staff ?

Coaches and medical staff receive real-time alerts from dashboards or mobile apps when critical conditions appairs , such as sustained high heart rates or fatigue indicators. This allows immediate action, improving player safety and optimizing performance during training or matches.

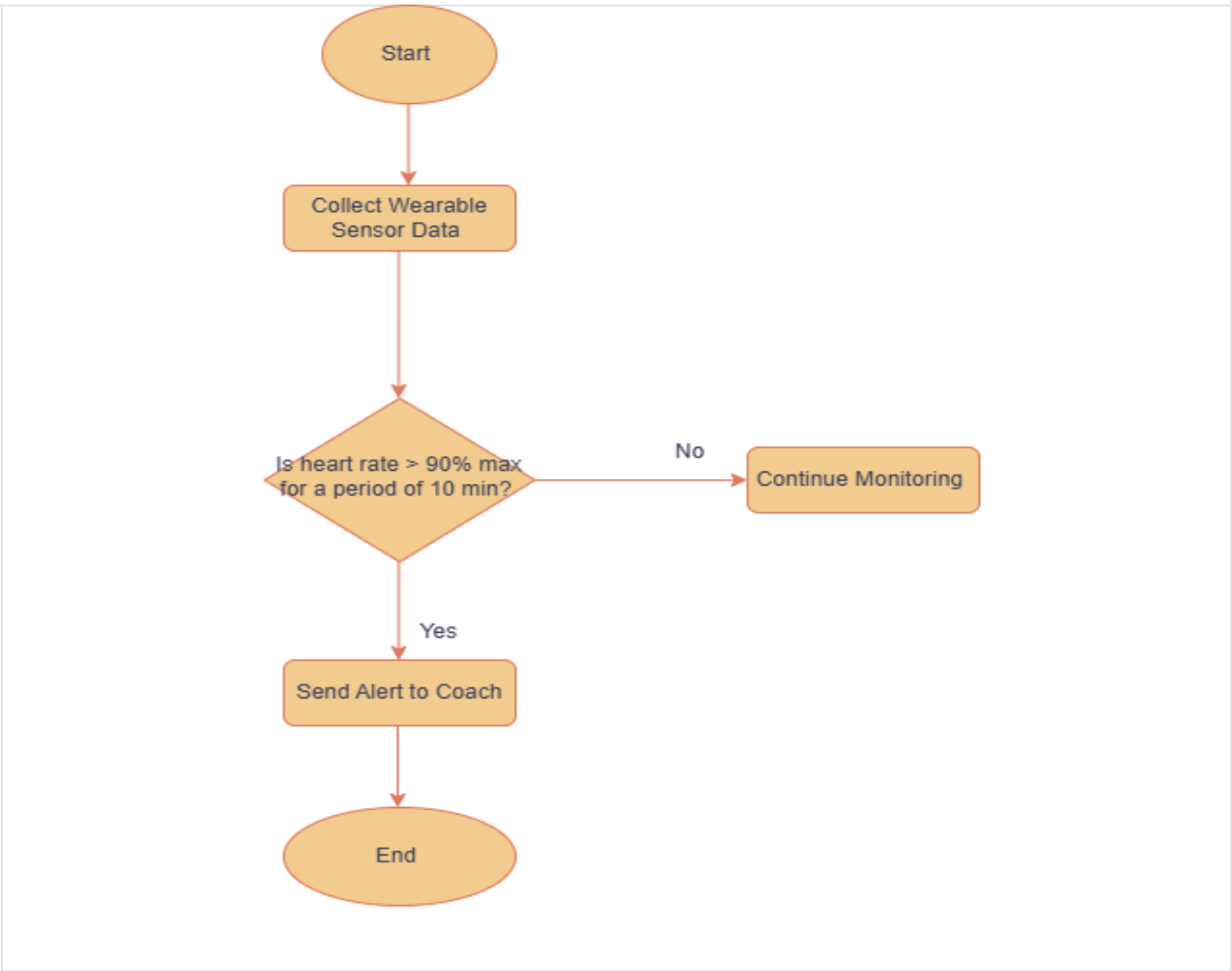


Fig 4 : A Simple Player Monitoring Flow Diagram

The Player Monitoring Flow Diagram explains how wearable sensors detect players physical performance in real time. The sensors collect data such as heart rate, movement. The system analyzes the data to identify critical conditions, like a sustained heart rate exceed 90% of the player's maximum for more than 10 minutes Period . If such a condition is detected, an alert is sent to the coaching staff or medical team to take action, such as substituting the player.

IV. Potential Use Cases :

1. **Real-Time Field Monitoring:** The system enables real-time monitoring of field conditions, allowing groundskeepers to identify issues such as dry spots, compaction, or excessive moisture levels and take immediate corrective actions.

2. **Injury Prevention and Safety:** By analyzing player movement punctures and turf automatically, the system can actively spot injury risk factors like uneven grass territory or slippery regions and enforce measures to abate these hazards.

3. **Optimized Maintenance Scheduling:** the prediction of turf deterioration allows to optimally time the turf maintenance activities (such as irrigation, fertilizing, aerating) leading to minimal disturbance of field usage while maximizing the quality of the playing surface.

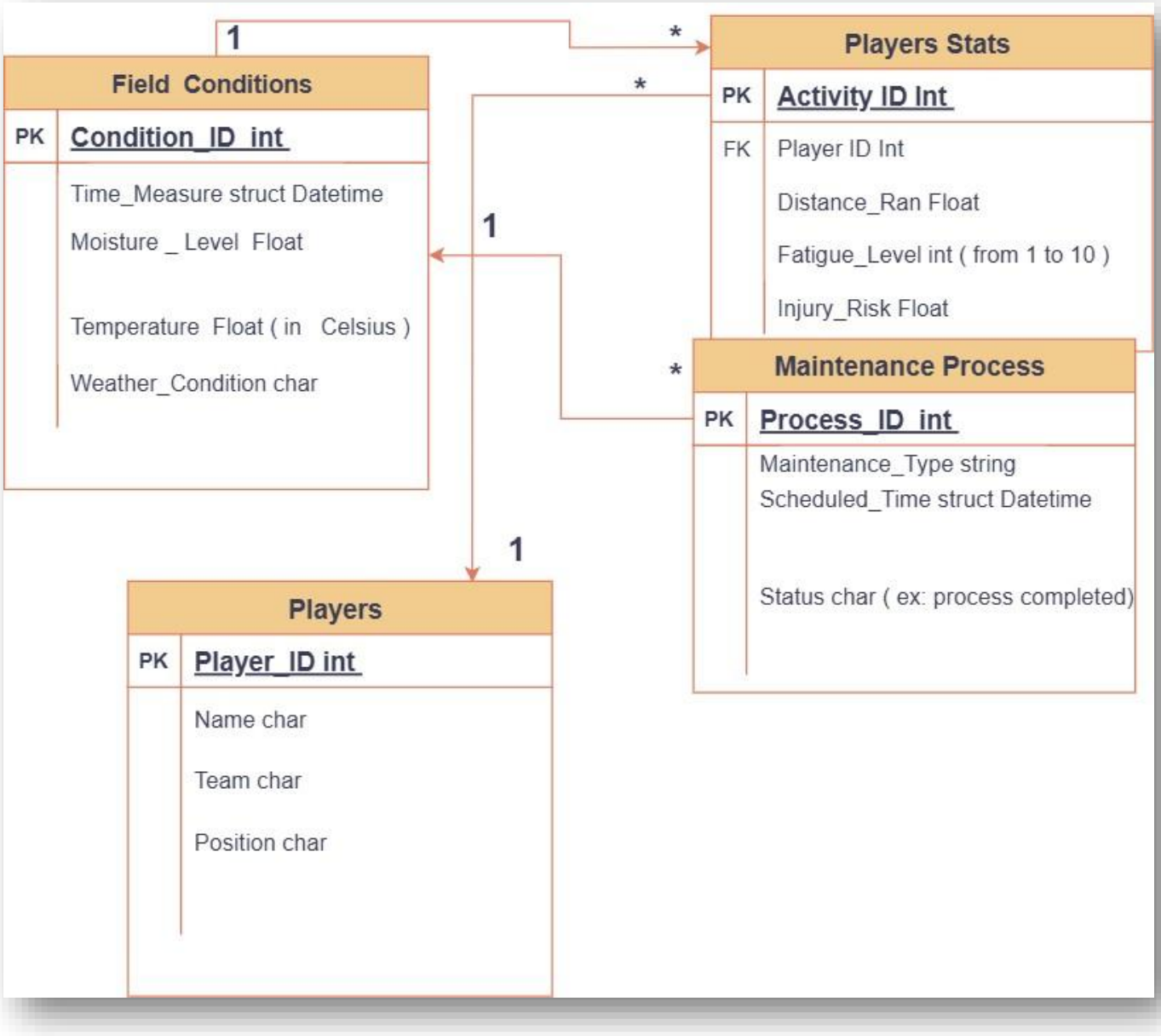


Fig 5: Potential ER Diagram for The IOT System

The ER Diagram of the Smart Management System for an efficient Football Pitch represents the relationships between key entities: **Field Conditions**, **Players**, **Player Stats**, and **Maintenance Process.** Each entity is designed to capture the necessary data for real-time monitoring, efficient maintenance, and player safety enhancement.

The Entities and their Attributes:

1.Field Conditions: Captures the real-time data from IoT sensors about the pitch, including attributes like Condition_ID, Moisture_Level, Temperature, and Weather condition .

2.Players: Represents players participating in games or training, with attributes like Player_ID, Player_Name, Team and Position

3.Players Stats: it contains the performance metrics such as Activity ID , Fatigue_Level, Distance_Covered and inury risk collected from wearable devices, and uses Player_ID as a foreign key to link with the **Players** entity for personalized insights .

4.Maintenance Process : it records actions performed to maintain the pitch, like irrigation or aeration. Attributes include Process _ID, Maintenance Type, Status and Scheduled Time , it enables the tracking of responses to specific field conditions.

Relationships: A one-to-many relationship between Field Conditions and Maintenance Process where each field condition can result in multiple maintenance actions (e.g., aeration or irrigation triggered by dry soil).

A one-to-many relationship by the players and Player stat as a single player can have multiple performance records over time.
A one -to many relationship that Field conditions and Player Stat connects player performance to specific field conditions, offering insights into how the pitch affects gameplay or safety.
After briefly describing the ERD design, the paper talks about the implementation of the database system using Python and SQL. This model allows for flexibility to scale, as new attributes or entities, such as adding sensors or including player information in case of Accenture, can be added without designing the database from scratch. The Model also establishes a compatibility with edge databases for near real-time decisions and cloud storage for historical analysis . So it also insight Generation by making Relationships between entities (e.g., field conditions affecting player stats) enabled and advanced analytics, such as predicting injury risks or optimizing maintenance schedules.

V. Performance Analysis and Cost Evaluation for The IoT-System

Throughput Experiment for the IOT smart system for an efficient football field : Throughput refers to the amount of data the system can process or transmit over a given period. For the IoT system, this involves measuring Data Collection Rate from Sensors , Transmition Rate to the Edge Gateway and Cloud , Processing Rate at the Edge and Cloud .
So we can use it as an example : we need to deploy a representative subset of sensors (e.g., 5 turf sensors, 2 weather sensors, and 5 wearables) and we Simulate real-world conditions with varying data rates (e.g., every 5 seconds, 2 minute).
And in order to measure it we also have to track the data packets generated per second.Measure the time it takes for data to travel from sensors to the cloud and be processed and also to forget to record throughput as **data per second (bps).**

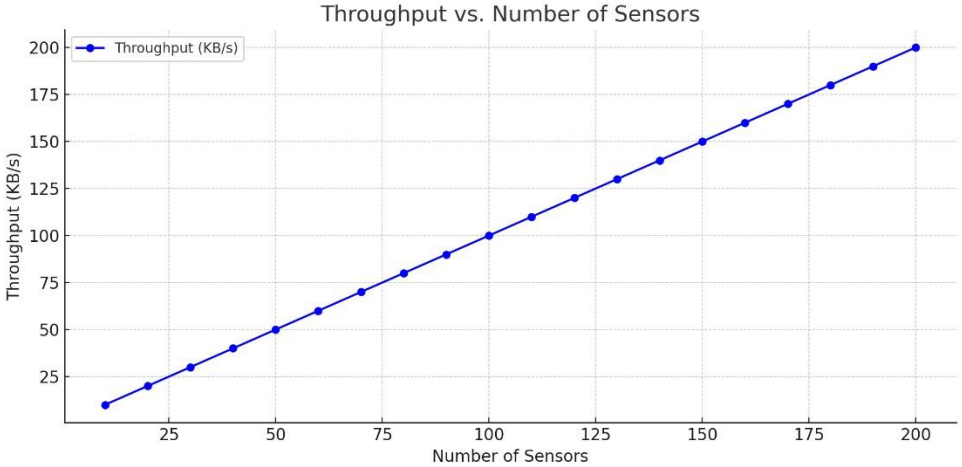


Fig 6 : line graph Throughput (KB/s) vs. Number of Sensors for the smart management system of a football field

Here is a line graph illustrating **Throughput (KB/s) vs. Number of Sensors** for the smart management system of a football field , The graph shows a linear relationship, with throughput increasing proportionally to the number of sensors deployed.

Cost Estimation for Cloud Storage:

So let's talk now about the cost estimation for the cloud storage of my IOT Smart management System of the efficient Football Field so as assumptions we can take into account that via a lot of reports : so for example let's say for the Cloud

Storage : $0.023 per GB per month (storage) and for the Data Growth Each sensor generates 1.44 MB/day. Data stored for 30 days (1 month).
Total sensors: 100,200 or 500 sensors .

Calculations :

Data Generated per Month:

For N sensors:

Monthly Data (GB)=N×1.44MB/day×30days÷1024MB/GB

Storage Cost:

Storage cost for N sensors : Storage Cost (USD)=Monthly Data (GB)×0.023 (0.023 refers to **$0.023/GB**: This is a standard rate charged by many cloud storage providers (e.g., AWS S3, Azure, Google Cloud Storage) for storing 1 GB of data for one month. It applies to the **entire data stored** for the month. For example, if you store 10 GB for a month, the cost is 10×0.023=0.23 USD

Retrieval Cost:

Retrieval Cost (USD)=Monthly Data (GB)×0.1×0.01 ($**0.01/GB**: This is the cost to retrieve 1 GB of data from cloud storage) and for **10% (0.1)**: This represents the assumption that only 10% of the stored data will be retrieved in a given month

Total Cost: Total Cost (USD)=Storage Cost +Retrieval Cost

So, for example for 100 sensors the calculations below: The Monthly Data:
100 x 1.44 x30 /1024 = 4.22 GB
Storage Cost: 4.22 x 0.023 = 0.097 USD
Retrieval Cost: 4.22 x 0.01 x0.1 =0.0042 USD
Total Cost: 0.097+0.0042=0.1012 USD

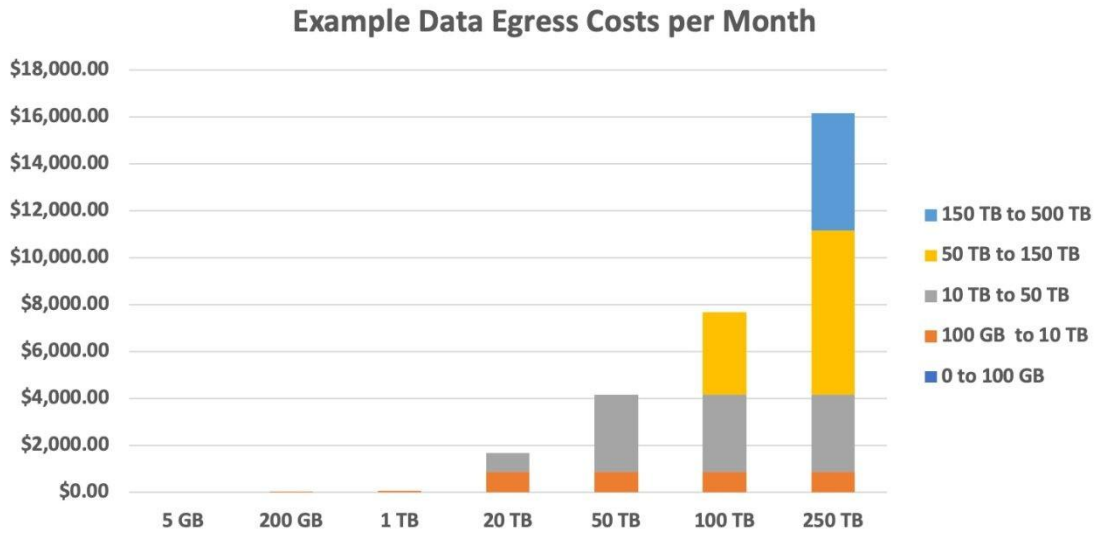


Fig 7 : Data Cost per Month for the IOT System

VI. CONCLUSION

This project highlights the innovative way that IoT technology can be used to enhance sports facilities with a New IoT-Smart Management System for Efficient Football Pitch in Field Management. They are using this system to monitor turf via sensors, edge computing and cloud analytics to capture real-time information about turf conditions, player performance and weather data. Not only does this help retain pitches and protect the players on-site, it also improves the conditions of play and extends the life of the field. So Overall, this IoT system is a game-changer for sports field management, ushering in a new era of data-driven decision-making for smarter, safer, and more sustainable sports environments. As the system evolves and is developed further, it could change the landscape of sports management, but also expand into other domains whilst paving the way for the use of technology in sports.

References :

[1] Google Chrome https://www.google.de/

A. **[2] /upperhand.com/football-club-management-system // Upperhand The best system to manage your Football Club /**

B. **Wikipedia official website**

C. **ChatGPT //**

[3] https://github.com/huytag/Mini-Football-Field-Management-System //

[4] https://www.drawio.com/

[5] G. Gupta & N. Meena. (2018). "IoT-Based Smart Sports Field Maintenance Systems." ***International Journal of Engineering and Technology (IJET).***

[6] Baca, A., & Kornfeind, P. (2006). "Tactical Metrics for Soccer Using IoT." ***Journal of Sports Sciences.***

[7] D. Vasisht, Z. Kapetanovic, J. Won, X. Jin, R. Chandra, S.N. Sinha, A. Kapoor, M. Sudarshan, S. Stratman, "FarmBeats: An IoT platform for data-driven agriculture," In Proceedings of the 14th USENIX Symposium on Networked Systems Design and Implementation, Boston, USA, pp. 515–529, 27–29 March 2017.

[8] Holler, J., Tsiatsis, V., Mulligan, C., Avesand, S., & Karnouskos, S. (2014). ***From Machine-toMachine to the Internet of Things: Introduction to a New Age of Intelligence.***

[9] https://aws.amazon.com/s3/pricing/

[10] https://azure.microsoft.com/en-us/services/iot-hub/

[11] https://cloud.google.com/storage/pricing